Thesis Update #5

Milan Chhetri

Mohammad Ghamari

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Jacques 2013 - “Optical properties of biological tissues: a review” [1]

Paper #1 – Tissue Optical Properties

Overview:

- Review/analysis paper on tissue optics
- Provides general equations for absorption [$\mu_a(\lambda)$] and the reduced scattering coefficient [$\mu'_s(\lambda)$] rather than relying on tabulated values
- Anisotropy [$g(\lambda)$] and refractive index [$n(\lambda)$] are approximated from literature data
- Equations and approximations can be applied to a generic tissue with a known composition

Paper #1 – Tissue Optical Properties

Methods:

- Uses power law decay and Beer-Lambert absorption law to predict general form of $\mu_s'(\lambda)$ and $\mu_a(\lambda)$ respectively
- Specific form can be determined using an empirical fit of literature data on various tissues with known compositions

Paper #1 – Tissue Optical Properties

Results

- Optical properties of a generic tissue can be estimated using the following equations [1]
  - $\mu_s'(\lambda) = a * \left(\frac{\lambda}{500\ nm}\right)^{-b}$
    - Where μ_s' is the reduced scattering coefficent; a and b are tissue dependent parameters, within 400 – 1300 nm [1]
  - $\mu_s(\lambda) = \frac{\mu_s'(\lambda)}{1-g}$
    - Where μ_s is the scattering coefficient and g is the anisotropy factor [1]

Paper #1 – Tissue Optical Properties

Results

- $\mu_a(\lambda) = BS[\,\mu_{a,HbO_2}(\lambda)\,] + B(1-S)[\,\mu_{a,Hb}(\lambda)\,] + W[\,\mu_{a,water}(\lambda)\,] + F[\,\mu_{a,fat}(\lambda)\,] + M[\,\mu_{a,melanin}(\lambda)\,]$
  - Where μ_a is the absorption coefficient for tissue; $B, S, W, F, and\ M$ are the tissue blood, oxygen, water, fat, and melanin content respectively [1]
- $n_{tissue} = (1-W)n_{dry} + Wn_{water}$
  - Where n is the refractive index and W is the tissue water content [1]
- a and b parameters for $\mu_s'(\lambda)$ parameters can be determined using literature data
- $B, S, W, F, and\ M$ provides by Jacques 2013 for different types of tissue, but absorption spectra provided by other sources[6][7][8][9]

Paper #1 – $\mu_s'(\lambda)$ for Three-Layer Skin

Layer	Source row	$a(cm^{-1})$	b
Epidermis	[1]	68.7	1.161
Dermis	[1]	45.3	1.292
Subcutaneous fat	[1]	13.7	0.385

Paper #1 - $\mu_s'(\lambda)$ for Three-Layer Skin

- Epidermis
  - $\mu_s'(\lambda)[cm^{-1}] = 68.7 * \left(\frac{\lambda}{500\ nm}\right)^{-1.161}$
- Dermis
  - $\mu_s'(\lambda)[cm^{-1}] = 45.3 * \left(\frac{\lambda}{500\ nm}\right)^{-1.292}$
- Subcutaneous Tissue
  - $\mu_s'(\lambda)[cm^{-1}] = 13.7 * \left(\frac{\lambda}{500\ nm}\right)^{-0.385}$

Paper #1 – $\mu_a(\lambda)$ for Three-Layer Skin

- $\mu_a(\lambda) = BS[\,\mu_{a,HbO_2}(\lambda)\,] + B(1-S)[\,\mu_{a,Hb}(\lambda)\,] + W[\,\mu_{a,water}(\lambda)\,] + F[\,\mu_{a,fat}(\lambda)\,] + M[\,\mu_{a,melanin}(\lambda)\,]$
  - Will need to compile literature data for the absorption spectra of each tissue constituent as well as tissue composition for each skin layer [1][6-9]

Paper #1 – $\mu_a(\lambda)$ for Tissue Constituents

Constituent absorption spectra:

- $\mu_{a,melanin}(\lambda) = 688.7\left(\frac{\lambda}{500\ nm}\right)^{-3.48}$ [8]
- $\mu_{a,Hb\ \ _2}(\lambda),\ \mu_{a,Hb}(\lambda), \mu_{a,water}(\lambda),\ \mu_{a,fat}(\lambda)$ → TBD [6][7][9]

Skin Layer Composition:

- Epidermis [1]
  - M=0.025 ; B,S,W,F = TBD
- Dermis [1]
  - M=0 ; B,S,W,F = TBD
- Subcutaneous Tissue [1]
  - M=0 ; B,S,W,F = TBD

Takeaways

- Need to account for glucose effects on tissue optics using Ge 2024
- Ge 2024 used Iino 2003 results as a baseline three-layer tissue model so Iino 2003 will be reviewed first

Iino 2023 - “Monte Carlo simulation of near infrared reflectance spectroscopy in the wavelength range from 1000 nm to 1900 nm” [2]

Paper #2 – Monte Carlo Simulation

Overview:

- Studied NIR propagation using a three-layer tissue model from literature data
- Photon diffusion is inaccurate for short SDS → use Monte Carlo for photon transport model
- Used a detector limit in simulation to remove very low-energy photons, which hidden under noise under real test conditions
- Validated simulation results with liquid phantom measurements using detector limit

Results

- Used g=0.9 and n=1.37 for each layer
- Calculated for a three-layer tissue model
  - Average photon visit depth
  - Photon energy fraction absorbed by layer
  - Average photon path length
  - Propagation path
- At 1600 nm
  - As SDS increased, dermis contribution increased, but reflected intensity decreased
- Baseline three-layer tissue model used by Ge 2024

Ge 2024 - “Evaluation and Validation on Sensitivity of Near-Infrared Diffuse Reflectance in Non-Invasive Human Blood Glucose Measurement” [3]

Paper #3 – Glucose-Dependent Tissue Optics

Overview:

- Analyzed how tissue optics changed with respect to both wavelength and tissue concentration
- Paper found that scattering presented the most dominant dependence on glucose concentration
- Uses absorbance instead of reflected intensity
- Sensitivity of absorbance w.r.t glucose ($\frac{dA}{dC_g}$) largest at 1550 nm

Paper #3 – Glucose-Dependent Tissue Optics

Methods:

- Used three-layer tissue model for MCXtreme Simulation [2]
  - Epidermis: 0.1 mm
  - Dermis: 1 mm
  - Subcutaneous Tissue: 10 mm
- $\frac{d\mu_s/dC_g}{\mu_s} = -0.29\%\ per\ mmol/L$ [3]
- $\mu_s(\lambda, C_g) = \mu_{s0}(\lambda)[1 - 0.0029(C_g - C_{g0})]$
- Applies mainly to wavelengths within 1000-1700nm

Takeaways

- Need equations that describe three-layer tissue optics dependence on both wavelength and glucose
- Ge 2024 provides the effect of glucose on three-layer tissue optics, but only for 1000 – 1700 nm [3]
- Use equations provided by Jacques 2013 and other literature data to create a wavelength-dependent tissue optics model for 400 – 1300 nm ($\mu_a(\lambda)$, $\mu_s'(\lambda)$, $g(\lambda)$, $n(\lambda)$)
- Compare with Iino 2003 three-layer tissue optics model for 1000 – 1900 nm [2]
- Add glucose effects on absorption and scattering using Ge 2024 findings to obtain final equations ($\mu_a(\lambda, C_g)$, $\mu_s'(\lambda, C_g)$)

Takeaways

- Final equations [$\mu_a(\lambda, C_g)$, $\mu_s'(\lambda, C_g)$, $g(\lambda)$, $n(\lambda)$] will be categorized into confidence tiers based on the overlap of literature ranges
  - 1st : 1000 – 1300nm
    - Strongest support in both glucose and wavelength dependence from Ge 2024, Jacques 2013, and Iino 2003
  - 2nd : 1300 – 1700nm
    - Supported by Ge 2024 for glucose dependence, but outside Jacques 2013 validated range for scattering wavelength dependence
  - 3rd : 400-1000nm
    - Supported by Jacques 2013 for wavelength dependence but outside Ge 2024 validated range for glucose dependence
- May need to model final equations as piecewise functions once more data is obtained

Takeaways

Once we have a baseline three-layer tissue model whose optics depend on both glucose concentration and wavelength, we can compare it with other reflectance mode sensor designs

Liu 2025 - “Design Key Points of High-Performance Diffuse Reflectance Optical Sensors for Non-Invasive Blood Glucose Measurement” [4]

Paper #4 – Sensor Configuration

Overview:

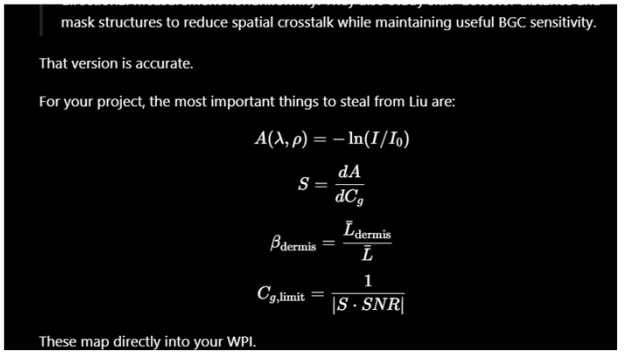
- Used seven-layer tissue model and obtained tissue optics from the literature
- Used three metrics to determine optimal reflectance mode sensor configuration
  - Depth Resolution
  - Detection SNR
  - Human-Sensor Interface Coupling

Paper #4

Methods:

- $SNR = \left|\frac{I}{\sigma_I}\right|$
  - Where I is the average reflected intensity and σ is the standard deviation of the light intensity over a time window
- $A(\lambda, \rho) = -\ln(\frac{I(\lambda,\rho)}{I_0(\lambda)})$
- $C_{g,limit} = \frac{1}{|S*SNR|}$
  - Where $S = \frac{dA(\lambda,\rho)}{dC_g}$
  - This limit quantifies how much noise can be fed into a detectors before gluscose change become undetectable
- Good Depth Resolution → focuses on NIR penetration in the dermis since blood vessels are heavily concentrated there

Paper #4



Methods:

- In MCeXtreme
  - SDD (skin-detector distance) – air gap
  - Test 2mm, 4mm, and 6mm for SDS
  - Detector shape and size (patch or ring, patch radius, ring radius, ring width)
  - Varied interface mode
- Ring detector configuration reduce anisotopic effects associated with human skin (Ex: uneven skin surface)
- Dual detector configuration supported through experimental results and by Ge 2024 and Han 2022 [4][3][5]

Paper #4

Results

- SDS controls tissue depth
- Ring-shaped InGaAs detectors performed better compared to circular detectors
- Air gap improves wearbility but causes large light energy loss around 85%
- Differential SDS measurements improved stability

To-do

- Use literature data to estimate $u_{a,tissue}(\lambda)$ for three-layer model and Ge 2024 to model glucose effects on model
- Use tissue optics equations to model wavelengths and glucose effects in MCeXtreme
- Find more literature on glucose/wavelength-dependent tissue optics for a broader range of wavelengths

References

Main References:

[1] S. L. Jacques, “Optical properties of biological tissues: a review,” *Phys. Med. Biol.*, vol. 58, no. 11, pp. R37–R61, May 2013, doi: 10.1088/0031-9155/58/11/R37.

[2] K. Iino, K. Maruo, H. Arimoto, K. Hyodo, T. Nakatani, and Y. Yamada, “Monte Carlo simulation of near infrared reflectance spectroscopy in the wavelength range from 1000 nm to 1900 nm,” *Opt. Rev.*, vol. 10, no. 6, pp. 600–606, 2003.

[3] Q. Ge, T. Han, R. Liu, Z. Zhang, D. Sun, J. Liu, and K. Xu, “Evaluation and Validation on Sensitivity of Near-Infrared Diffuse Reflectance in Non-Invasive Human Blood Glucose Measurement,” *Sensors*, vol. 24, no. 18, Art. no. 5879, Sep. 2024, doi: 10.3390/s24185879.

[4] W. Liu, T. Han, W. Chen, J. Chen, Q. Ge, D. Sun, J. Liu, and K. Xu, “Design Key Points of High-Performance Diffuse Reflectance Optical Sensors for Non-Invasive Blood Glucose Measurement,” *Sensors*, vol. 25, no. 4, Art. no. 998, Feb. 2025, doi: 10.3390/s25040998.

Supporting References:

[5] Han, T.; Liu, J.; Liu, R.; Chen, W.; Yao, M.; Liu, X.; Ge, Q.; Zhang, Z.; Li, C.; Wang, Y.; et al. In vivo near-infrared noninvasive glucose measurement and detection in humans. Appl. Spectrosc. 2022, 76, 1100–1111.

References

Supporting Papers:
[6] Water Absorption Spectra:
https://omlc.org/spectra/water/abs/index.html
[7] Hemoglobin Absorption Spectra:
https://omlc.org/spectra/hemoglobin/
[8] Melanin Absorption Spectra:
https://omlc.org/spectra/melanin/mua.html
[9] Fat Absorption Spectra:
https://omlc.org/spectra/fat/